

ISI-MAC — channel, decoder, capacity

Two-user multiple access channel with deterministic intersymbol interference (ISI).
Two transmitters encode polar codewords $x^N, y^N \in \{0, 1\}^N$ and BPSK-modulate.

Channel model

$$Z_i = (1 - 2X_i) + (1 - 2Y_i) + h[(1 - 2X_{i-1}) + (1 - 2Y_{i-1})] + W_i$$

$$W_i \sim \mathcal{N}(0, \sigma^2) \text{ i.i.d.}, \quad h = 0.3 \text{ (tap)}, \quad \sigma^2 = 10^{-0.6} \text{ (SNR = 6 dB)}.$$

Channel state

$$S_i = (X_{i-1}, Y_{i-1}) \in \{0, 1\}^2, \quad |\mathcal{S}| = 4. \quad \text{Transition is deterministic: } S_{i+1} = (X_i, Y_i).$$

Analytical decoder (SCT) — joint forward-backward + computational-graph SC

Forward and backward recursions on $\mathcal{S}$:

$$\alpha_t(s) = \log \sum_{x, y, s_-} e^{\alpha_{t-1}(s_-)} W(z_t | x, y, s_-, s) \cdot P(x)P(y)$$

$$\beta_t(s) = \log \sum_{x, y, s_+} W(z_{t+1} | x, y, s, s_+) e^{\beta_{t+1}(s_+)} P(x)P(y)$$

Per-position joint marginal:

$$\log P(z_1^N, X_t = x, Y_t = y) = \text{LSE}_{s_-, s_+} [\alpha_{t-1}(s_-) + \log W(z_t | x, y, s_-, s_+) + \beta_{t+1}(s_+)]$$

The per-position (2×2) marginals feed a Ren-et-al-style computational-graph SC tree that decodes along an arbitrary path through the rate region. Class C corner-rate (all U then all V).

Neural decoder (NPD) — chained corner-rate

Stage 1 (U): BiGRU over z_1^N produces per-position embeddings; neural CheckNode/BitNode/Emb2LLR

tree decodes U . Stage 2 (V): same architecture with $\hat{X}$ as extra per-position side info.

Trained rate-1 (all positions information), then MI per synthesized channel $W_N^{(i)}$ selects $\mathcal{A}_U, \mathcal{A}_V$.

Symmetric capacity and rate

$I(W) = I(X, Y; Z)$ under uniform X, Y . Class C corner-rate uses

$$R_U = I(X; Z), \quad R_V = I(Y; Z | X), \quad \text{total throughput } R = (R_U + R_V)/2.$$

ISI-MAC

$\eta = 0.3$, SNR = 6 dB. Chained Class C corner-rate. Each decoder uses its own MC-derived frozen design.

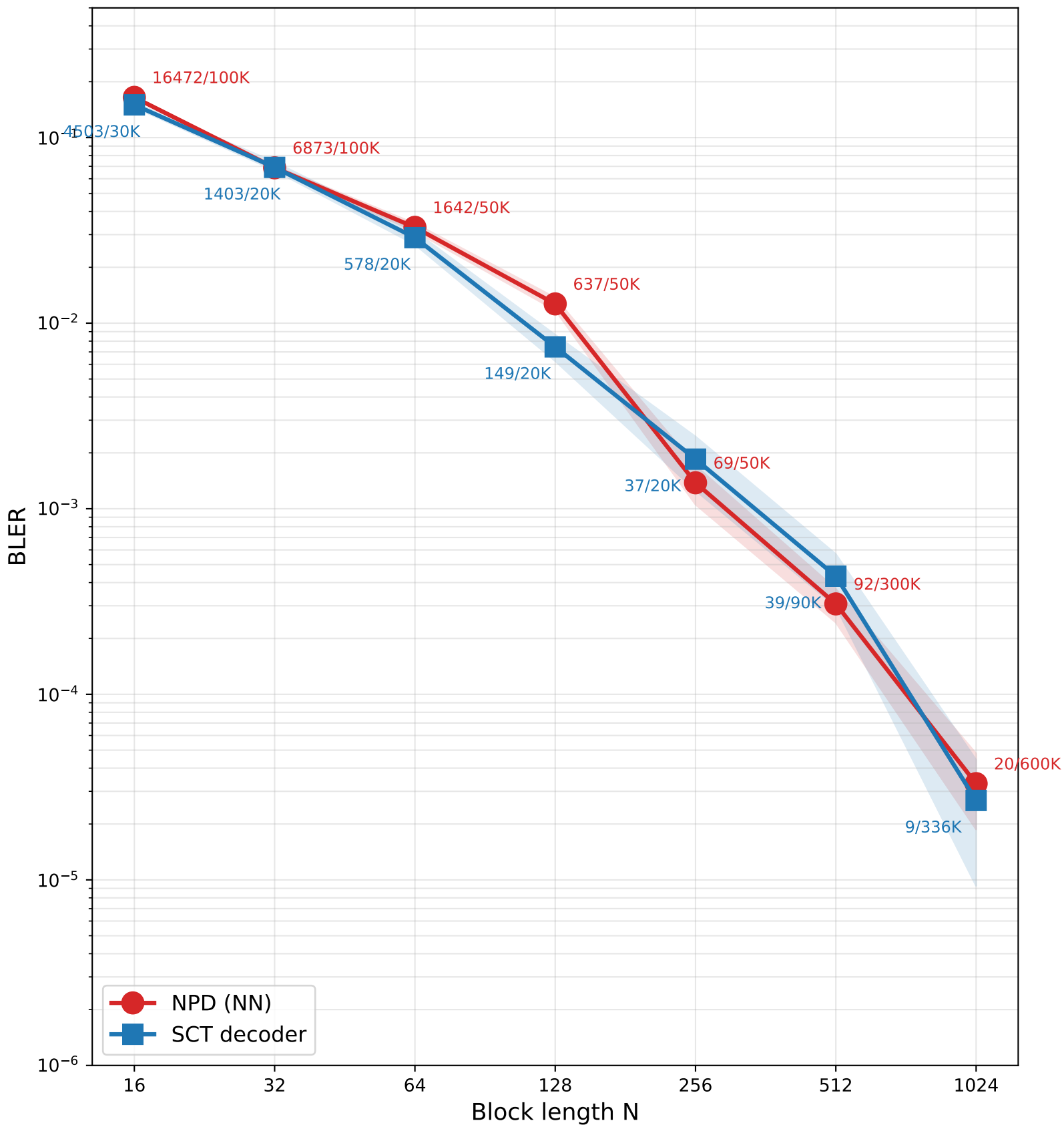
N	NPD BLER	NPD errs/CW	analytical BLER	analytical errs/CW	NPD/anal
16	1.647e-01	16,472/100,000	1.501e-01	4,503/30,000	1.10
32	6.873e-02	6,873/100,000	6.910e-02	1,403/20,000	0.99
64	3.284e-02	1,642/50,000	2.890e-02	578/20,000	1.14
128	1.270e-02	637/50,000	7.450e-03	149/20,000	1.70
256	1.380e-03	69/50,000	1.850e-03	37/20,000	0.75
512	3.070e-04	92/300,000	4.330e-04	39/90,000	0.71
1024	3.300e-05	20/600,000	2.680e-05	9/336,000	1.23

Notes

- NPD: BiGRU z-encoder + neural CheckNode/BitNode tree. d=16 hidden=100 unless retrained larger at small N.
- Analytical: joint-trellis SC (SCT) on $\mathcal{S}=4$ MAC lattice, MC design at 3 dB.
- "NPD/anal" ratio < 1 $\Rightarrow$ NPD better (green); > 1.1 $\Rightarrow$ analytical better (red).
- All BLERs are chained corner-rate (Class C) at target rate. SNR = 6 dB.
- Most points have ≥ 30 block errors $\rightarrow$ tight 95% Poisson CIs.

ISI-MAC

$h = 0.3, \text{SNR} = 6 \text{ dB}$



MA-AGN-MAC — channel, decoder, capacity

Two-user MAC with moving-average additive Gaussian noise. The noise process is AR(1) — the noise state is a continuous real number, so no finite-state trellis applies directly.

Channel model

$$Z_i = (1 - 2X_i) + (1 - 2Y_i) + N_i, N_i = \alpha N_{i-1} + W_i$$

$$W_i \sim \mathcal{N}(0, \sigma^2(1 - \alpha^2)) \text{ i.i.d.}, \text{Var}(N_i) = \sigma^2 \text{ (stationary)}, \alpha = 0.5, \text{SNR} = 6 \text{ dB}.$$

Why no off-the-shelf SCT

The noise carries an infinite-precision real state N_{i-1} across positions. A naïve trellis on the noise state has $|\mathcal{S}| = \infty$. Aharoni et al. 2024 state "no available SC decoding rule for channels with continuous state space" and only show NPD curves for MA-AGN in their Fig. 4.

Memoryless SC baseline (used in this report)

Treat the AR(1) noise as i.i.d. Gaussian with variance σ^2 .

$$\text{Stage 1 LLR (V marginalized uniform): } \text{LLR}_U(z) = \log \frac{\mathcal{N}(z; 2, \sigma^2) + \mathcal{N}(z; 0, \sigma^2)}{\mathcal{N}(z; 0, \sigma^2) + \mathcal{N}(z; -2, \sigma^2)}$$

$$\text{Stage 2 LLR (given } \hat{x}_i\text{): } \text{LLR}_V(z; \hat{x}_i) = \frac{(z - m_0)^2 - (z - m_1)^2}{2\sigma^2}, m_v = (1 - 2\hat{x}_i) + (1 - 2v).$$

Whitened-SCT baseline (stronger; running in our work, not in the paper)

Pre-filter $z'_i = z_i - \alpha z_{i-1}$. Then

$$z'_i = [(1 - 2X_i) - \alpha(1 - 2X_{i-1})] + [(1 - 2Y_i) - \alpha(1 - 2Y_{i-1})] + W_i$$

with $W_i \sim \mathcal{N}(0, \sigma^2(1 - \alpha^2))$ i.i.d. — exactly ISI-MAC with $h = -\alpha$ and variance $\sigma^2(1 - \alpha^2)$.

Joint trellis SC on z' applies with $|\mathcal{S}| = 4$. This is the proper analytical baseline.

Neural decoder (NPD)

Same chained corner-rate NPD as ISI: BiGRU over z_1^N implicitly learns noise predictability.

Trained at rate-1; MI-based frozen design via fast-CE on synthesized channels $W_N^{(i)}$.

MA-AGN-MAC

$\alpha = 0.5$, SNR = 6 dB. Chained Class C corner-rate.

N	NPD BLER	NPD errs/CW	analytical BLER	analytical errs/CW	NPD/anal
16	1.465e-01	7,327/50,000	1.068e-01	5,339/50,000	1.37
32	9.925e-02	9,925/100,000	4.020e-02	2,010/50,000	2.47
64	2.650e-02	1,325/50,000	3.494e-02	1,747/50,000	0.76
128	6.180e-03	309/50,000	1.616e-02	808/50,000	0.38
256	6.800e-04	34/50,000	1.460e-03	73/50,000	0.47
512	5.170e-05	31/600,000	4.090e-04	41/100,000	0.13
1024	3.000e-05	30/1,050,000	1.430e-04	43/300,000	0.21

Notes

- NPD: BiGRU z-encoder + neural CheckNode/BitNode tree. d=16 hidden=100 unless retrained larger at small N.
- Analytical: memoryless SC ignoring noise correlation. Whitened-SCT (stronger) is running.
- "NPD/anal" ratio < 1 $\Rightarrow$ NPD better (green); > 1.1 $\Rightarrow$ analytical better (red).
- All BLERs are chained corner-rate (Class C) at target rate. SNR = 6 dB.
- Most points have ≥ 30 block errors $\rightarrow$ tight 95% Poisson CIs.

MA-AGN-MAC

$\alpha = 0.5, \text{SNR} = 6 \text{ dB}$

